

MID-TERM EXAMINATION
(B.Tech CSE AI/ECE AI/ AI&ML) (Semester 1st)
(October, 2023) OFF LINE mode

Subject Code: BAS 109

Time : 1 ½ Hours

Subject: Applied Mathematics

Maximum Marks : 30

Note: (a) Q. 1 is compulsory. Attempt any two questions from the rest.
(b) Use of calculator is not allowed.

Q1

(a) Define rank of a matrix. Find rank of the matrix

(2.5*4)

$$A = \begin{bmatrix} 3 & 5 & 2 \\ 4 & 2 & 3 \\ -1 & 2 & 4 \end{bmatrix}$$

(b) If λ is an eigenvalue of a matrix A then prove that $g(\lambda)$ is an eigenvalue of a matrix $g(A)$, where g is a polynomial.

(c) Find the inverse Laplace transform of $\tan^{-1}\left(\frac{2}{s^2}\right)$.

(d) Let V be a vector space. Let H and K be subspaces of V . Give an example to show that $H \cup K$ need not be a subspace.

Q2 (Attempt any Two Parts) UNIT-1

(5,5)

(a) Check whether the system is consistent or inconsistent. Give the complete solution set, if consistent.

$$\begin{aligned} 3x - 3y - 2z &= 23 \\ -6x + 4y + 3z &= -40 \\ -2x + y + z &= -12 \end{aligned}$$

(b) Find the inverse of a matrix $A = \begin{bmatrix} 6 & -3 & -2 \\ 4 & -1 & -2 \\ 10 & -5 & -3 \end{bmatrix}$ by using the Cayley-Hamilton theorem.

(c) Consider the matrix:

$$A = \begin{pmatrix} 7 & 1 & -1 \\ -11 & -3 & 2 \\ 18 & 2 & -4 \end{pmatrix}$$

Find the eigenvalues and the fundamental eigenvectors of A .

Q3 (Attempt any Two Parts) UNIT-2

(5,5)

(a) Find the Laplace transform of $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ e^x, & 1 < x \leq 4. \\ 0, & x > 4 \end{cases}$

(b) (i) Consider the vector space $M_{3 \times 3}(\mathbb{R})$ and the sets
 T_1 : the set of non singular 3×3 matrices
 T_2 : the set of singular 3×3 matrices.
Determine T_1 and T_2 are subspaces of $M_{3 \times 3}(\mathbb{R})$ or not?

(ii) Apply convolution theorem to find the inverse Laplace transform of $\frac{1}{(s-2)(s+3)}$.

(c) For the following linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 5 \\ -2 & 3 & -13 \\ 3 & -3 & 15 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix},$$

find bases for $\ker(T)$ and $\text{range}(T)$. Also, verify the Dimension theorem.